

Conda + MPI Setup for HPC Cluster

1. Install Miniconda

Download the Miniconda installer using `curl`:

```
curl -O https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
```

Install Miniconda:

```
bash Miniconda3-latest-Linux-x86_64.sh
```

Follow the installation prompts.

Initialize Conda:

```
~/miniconda3/bin/conda init bash
```

Reload the shell:

```
source ~/.bashrc
```

Verify:

```
conda --version
```

2. Create a Conda Environment with the Latest Python

```
conda create -n hpc_env python
```

Activate the environment:

```
conda activate hpc_env
```

Verify:

```
python --version  
which python
```

3. Install NumPy and mpi4py

```
conda install -c conda-forge numpy mpi4py
```

Verify:

```
python -c "import numpy; print(numpy.__version__)"
```

```
python -c "from mpi4py import MPI; print(MPI.Get_version())"
```

4. Load GCC and OpenMPI

```
module load compiler/gcc/12.3  
module load compiler/openmpi/4.1.5
```

Verify:

```
gcc --version  
mpirun --version
```

5. Create `example.py`

```
from mpi4py import MPI  
  
comm = MPI.COMM_WORLD  
  
rank = comm.Get_rank()  
size = comm.Get_size()  
  
if rank == 0:  
    print(f"[SENDER] Hello from rank {rank}")
```

```
print(f"Number of MPI processes: {size}")
else:
    print(f"    [RECEIVER] Rank {rank} is also running in the compute node")
```

6. Create `mpirun.sh`

```
#!/bin/bash

#SBATCH --job-name=my_mpi_job
#SBATCH --partition=cpu
#SBATCH --nodes=2
#SBATCH --ntasks=4
#SBATCH --time=00:05:00
#SBATCH --output=my_mpi_job_%j.out
#SBATCH --error=my_mpi_job_%j.err

# Load MPI
module load compiler/gcc/12.3
module load compiler/openmpi/4.1.5

# Initialize Conda
source ~/miniconda3/etc/profile.d/conda.sh

# Activate Conda environment
conda activate hpc_env

# Display Python information
echo "Python:"
which python
python --version

# Display MPI information
echo "MPI:"
which mpirun
mpirun --version

# Run the MPI program
mpirun -np $SLURM_NTASKS python example.py
```

If Miniconda was installed in your home directory using the instructions above, this should work:

```
source ~/miniconda3/etc/profile.d/conda.sh
```

7. Submit the Job

```
sbatch mpirun.sh
```

Example:

```
Submitted batch job 30021
```

8. Check Job Status

```
squeue -u $USER
```

9. View Output

Replace **30021** with your actual job ID:

```
cat my_mpi_job_30021.out
```

View errors:

```
cat my_mpi_job_30021.err
```

Expected output:

```
[SENDER] Hello from rank 0
Number of MPI processes: 4
[RECEIVER] Rank 1 is also running in the compute node
[RECEIVER] Rank 2 is also running in the compute node
[RECEIVER] Rank 3 is also running in the compute node
```

Conda needs to be activated **inside the Slurm batch script**, even if the Conda environment was activated on the login node.